

Spotify: What Makes a Song Popular?

Analysis of the Spotify Top 100, 2010-2019: SQL, Python, HTML/CSS/JavaScript

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Note: *personal project undertaken out of interest in data analysis applied to music. Data cleaning, modeling and querying in SQL, statistical analysis in Python, an interactive dashboard built with HTML/CSS/JavaScript, and a robustness test on a dataset 89x larger (part 7).*

1. Background and objective

Starting point of the project: a fictional record label wants to give its artists concrete recommendations on the characteristics of a song that make it popular, in order to maximize the chances of success of its upcoming releases. This is the business framework I set for myself to structure the analysis.

Question asked: which characteristics (genre, tempo, energy, danceability, artist configuration...) are associated with the most popular tracks in the Spotify Top 100 over the 2010-2019 decade?

Working hypotheses

- Musical genre strongly influences presence in the Top 100 and average popularity.
- The audio characteristics (energy, danceability, valence) of popular tracks change over time.
- Artist configuration (solo, duo, group) is associated with different levels of popularity.
- Audio characteristics alone can reasonably predict a track's popularity.

This last hypothesis is deliberately put to the test rather than assumed true: see sections 5 and 7.

2. Data and methodology

2.1 Data source

Kaggle dataset “*Top Spotify Songs from 2010-2019 by year*” (1000 tracks, 19 columns), enriched with an “artist_type” column (Solo / Duo / Trio / Group).

2.2 Data cleaning (Python)

Script: `python/01_clean_data.py`. Main steps:

- Renaming the source file's cryptic columns (L, r, rank, nrgy, dnce...) into explicit names (title, chart_position, artist, energy, danceability...).
- Typing numeric and date columns.
- Detecting and fixing a real data-quality issue: five purely numeric titles (“212”, “3005”, “22”...) and one title auto-corrupted by Excel: Beyoncé's “7/11”, automatically converted into a decimal fraction (0.636363...) by Excel. Without this fix, 6 rows would have been lost during cleaning.

- Feature engineering: release decade, popularity tier (Low/Mid/High/Very High), a “spoken content” proxy (high speechiness), detection of titles reappearing across multiple years.

Result: 1000 rows kept (0 loss), 23 columns, exported to data/spotify_clean.csv.

2.3 SQL database modeling

Script: python/02_build_db.py, builds a SQLite database in a star schema from the cleaned CSV (see the glossary, section 8, for an explanation of this term):

```
dim_artist(artist_id, artist_name, artist_type)      444 artists
dim_genre(genre_id, genre_name)                    132 genres
fact_songs(song_id, title, chart_year, ..., artist_id, genre_id) 1000 tracks
```

This choice (rather than a single flat table) makes it possible to write real SQL joins and produces a proper relational model, closer to how data is structured in production systems.

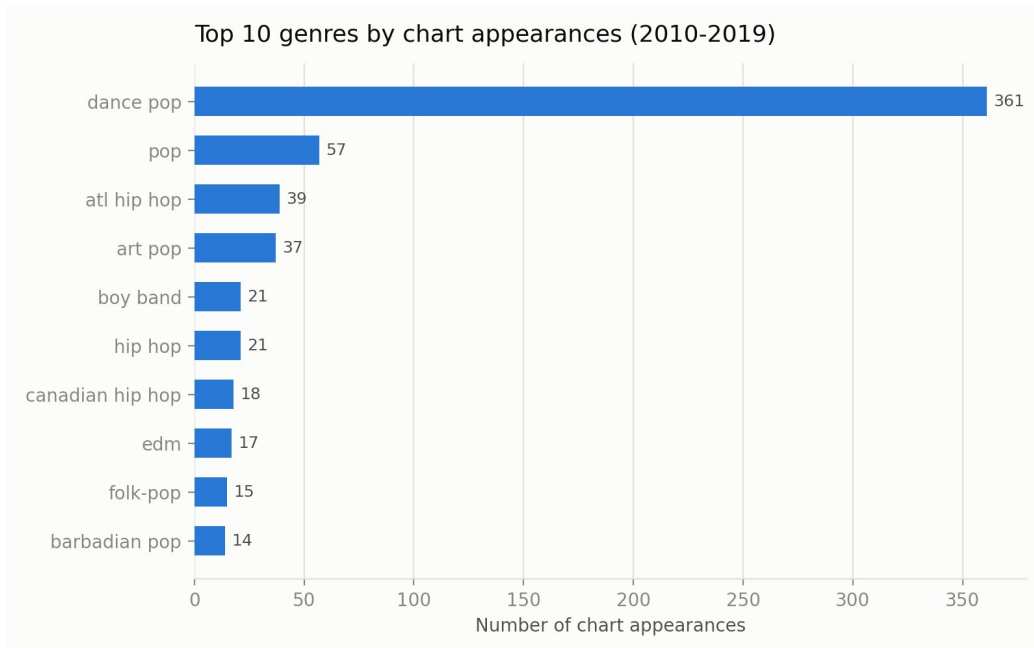
2.4 Analysis SQL queries

File: sql/analysis_queries.sql, 11 queries covering CTEs, window functions (ROW_NUMBER, LAG), aggregations, joins and subqueries. Representative excerpt (top 3 most popular tracks per year, a “top-N per group” technique frequently asked about in interviews):

```
WITH ranked AS (
  SELECT f.chart_year, f.title, a.artist_name, f.popularity,
         ROW_NUMBER() OVER (
           PARTITION BY f.chart_year ORDER BY f.popularity DESC
         ) AS rn
  FROM fact_songs f JOIN dim_artist a ON a.artist_id = f.artist_id
)
SELECT chart_year, title, artist_name, popularity
FROM ranked WHERE rn <= 3 ORDER BY chart_year, rn;
```

All queries were executed and validated (python/03_run_queries.py); the 11 result sets are exported to data/query_results/.

3. Key insights (Top 100, 2010-2019)



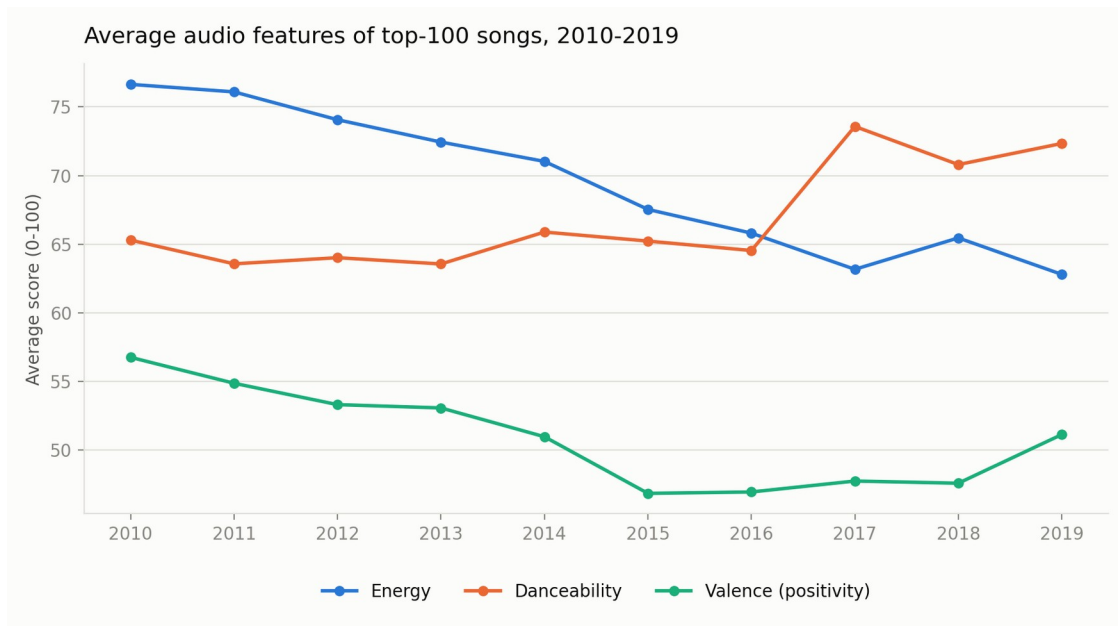
Top 10 genres by number of appearances in the Top 100 (2010-2019)

Dance pop dominates by a wide margin (36% of charted tracks), well ahead of pop (5.7%) and Atlanta hip hop (3.9%). A label betting on this genre mechanically maximizes its chances of chart presence; but section 7 shows that presence doesn't mean a better hit-conversion rate.

What characterizes pop?

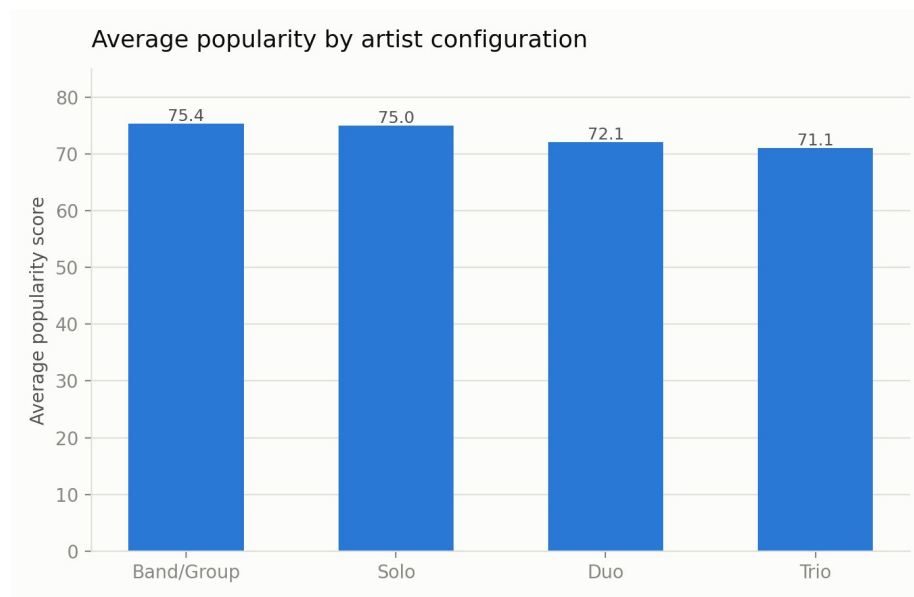
General context about the genre, not a finding drawn from this dataset: useful for understanding why dance pop, which belongs to this family, dominates the Top 100 rankings so heavily.

- **Catchy melodies:** easy to remember and sing along to.
- **Driving rhythm:** usually in 4/4 time, with a strong percussion presence.
- **Simple structures:** verse-chorus-bridge, which makes for easy listening.
- **Polished production:** electronic instruments, synthesizers, vocal effects.
- **Universal themes:** love, relationships, partying, personal emotions.



Evolution of average audio characteristics, 2010-2019

- **Energy declines steadily:** from 76.6 (2010) to 62.8 (2019), a nearly monotonic trend over 10 years.
- **Danceability jumps starting in 2017:** from ~65 to over 73, driven by the “tropical/dance-pop” wave of the late decade (Ed Sheeran, Dua Lipa, Post Malone...).
- **Valence (perceived positivity, see glossary) bottoms out in 2015-2016** before rising slightly again in 2019.

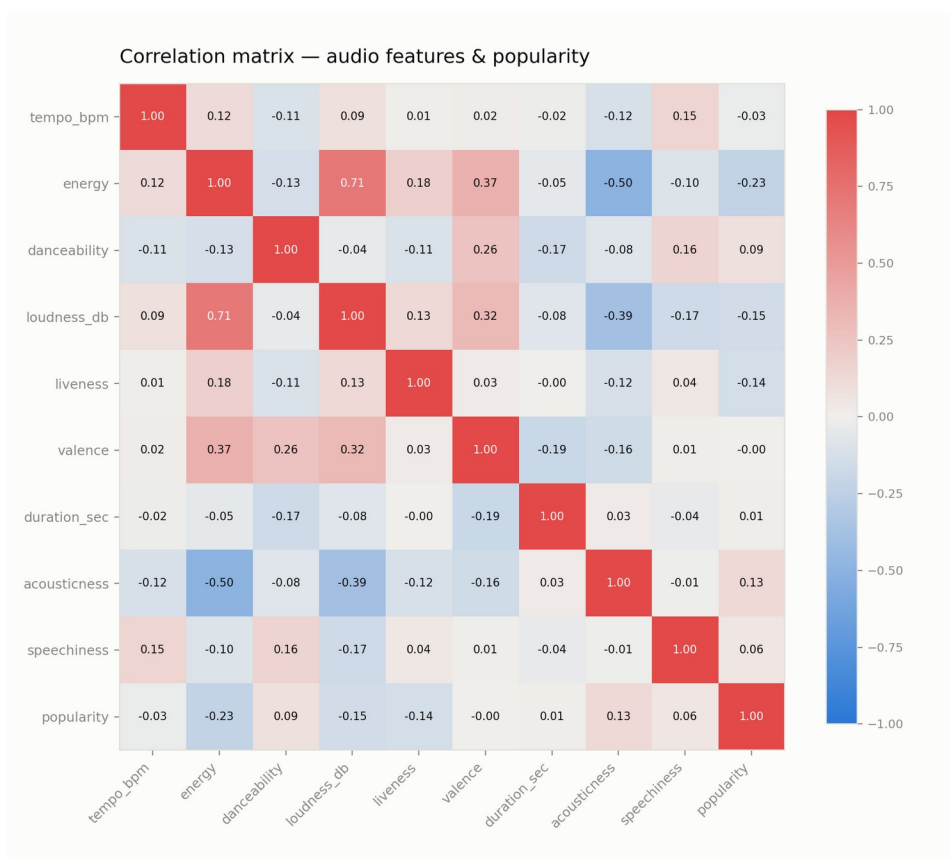


Average popularity by artist configuration

Groups and solo artists show nearly identical average popularity (75.4 vs 75.0); duos and trios lag slightly behind (72.1 and 71.1). The gap remains small; artist configuration is not a determining factor, contrary to the initial hypothesis.

4. Statistical analysis (Python)

Script: python/04_analysis.py (pandas + scikit-learn). Two parts: correlations between audio variables, then a linear regression to quantify what explains popularity.



Correlation matrix: audio characteristics and popularity

4.1 Validated correlations

Relationship	Correlation found	Initial hypothesis
Energy ↔ Loudness	0.71 (strong)	0.65: confirmed, similar magnitude
Danceability ↔ Valence	0.27 (moderate)	0.40: overestimated initially
Speechiness ↔ Popularity	0.06 (near zero)	“Explicit content has higher median popularity”: not testable with this proxy (see glossary: speechiness ≠ explicit content)

4.2 What actually predicts popularity?

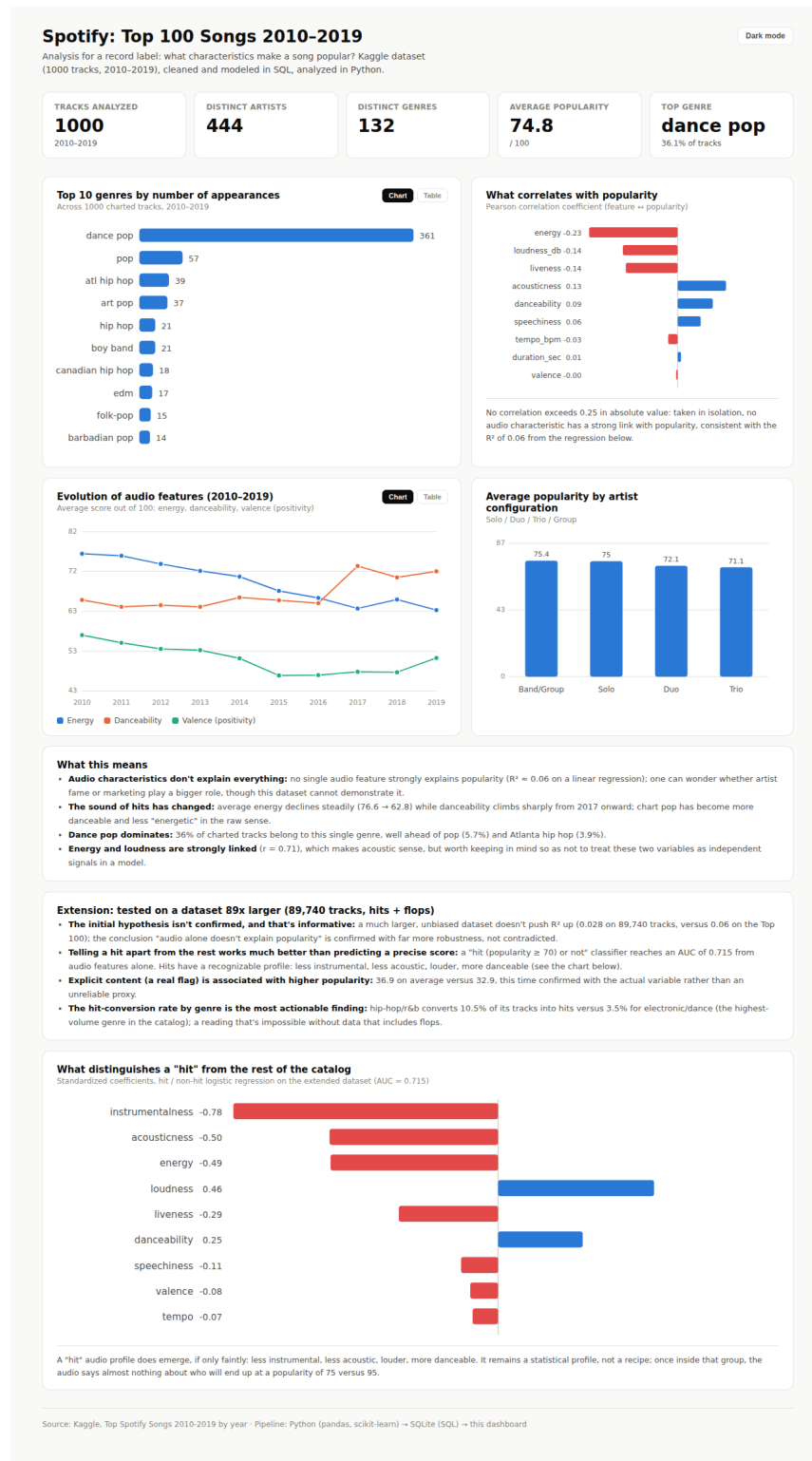
A linear regression (popularity ~ 9 audio characteristics, 80/20 train/test split) gives an R^2 of only 0.06 on the test set (see glossary for the definition of R^2): audio characteristics alone explain less than 6% of the variance in popularity.

- **This is the most important result of this first part:** contrary to the initial intuition, a track's “sound” doesn't explain much of its popularity. Section 7 tests whether this result holds on a much larger dataset.

This dataset contains no information about the artist (listener count, fame), promotion, or playlist placement; factors one can legitimately suspect play a role, though this project cannot demonstrate it. This is a limitation to keep in mind rather than a conclusion: see also sections 6 and 7.4.

5. Delivery: interactive dashboard

An interactive HTML dashboard (file `dashboard/spotify_dashboard.html`) brings together all the KPIs and charts above into a web-based interface, with a chart/table toggle and dark mode.



Preview of the interactive dashboard provided (`dashboard/spotify_dashboard.html`)

6. Recommendations for the label (based on the Top 100 alone)

- **Don't over-index on genre:** dance pop dominates by volume but doesn't have markedly higher individual popularity than other genres (74.0 on average, close to the overall median of 74.8).
- **The low R^2 invites questions about other levers:** if audio alone explains so little of popularity, one can wonder whether factors such as playlist placement or release strategy carry more weight than reworking a track's sound; a hypothesis this dataset cannot settle, for lack of data on the artist or promotion, but one worth digging into with the right data.
- **Follow the danceability trend:** the clear rise since 2017 suggests a market taste for more danceable tracks, even at the cost of reducing perceived raw energy.
- **Stay cautious with isolated correlations:** energy and loudness are so strongly linked ($r = 0.71$) that they shouldn't be treated as two independent levers in a product recommendation.

These recommendations are based on 1000 tracks that are all already hits (Top 100). The next section tests whether they hold on a sample 89x larger, which also includes tracks that never broke through.

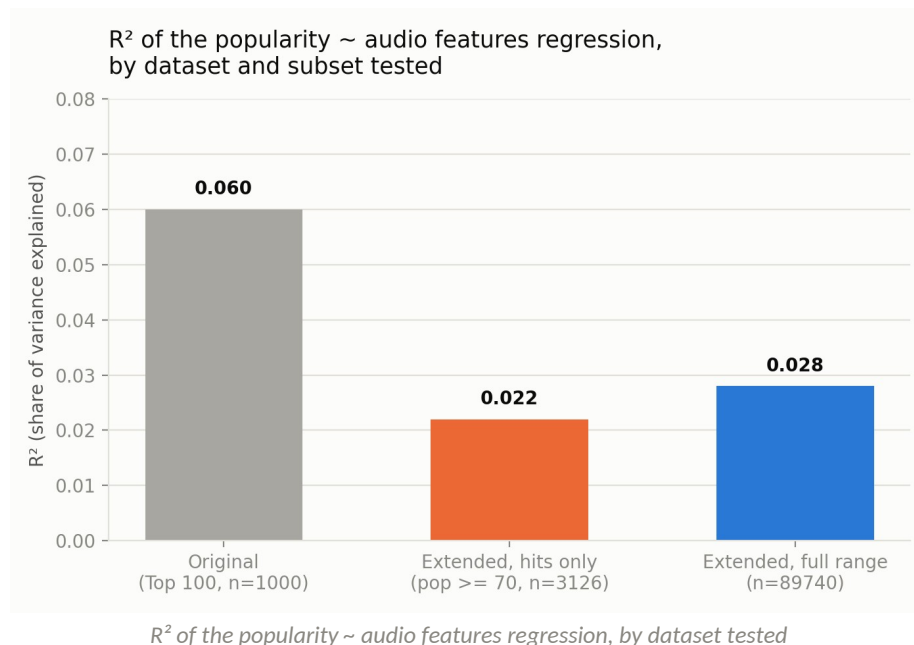
7. Going further: testing on a dataset 89x larger

Question raised after the first version of the analysis: would a larger dataset yield more conclusions? A tested answer, not an assumed one.

7.1 The real problem isn't volume, it's sample bias

The original dataset (1000 tracks) consists exclusively of hits (the Top 100 of each year): popularity there varies within a narrow window (60-100). This is a classic range-restriction bias in statistics, which mechanically compresses observable correlations. To test this, a second dataset was brought in: the “Spotify Tracks Dataset” (Kaggle, 114,000 tracks, 114 genres), which covers the entire catalog: hits AND flops, popularity from 0 to 100. After deduplication (a single track can be tagged under several genres), 89,740 unique tracks remain (script `python/06_clean_extended.py`).

7.2 The result is surprising, and it's reported as-is

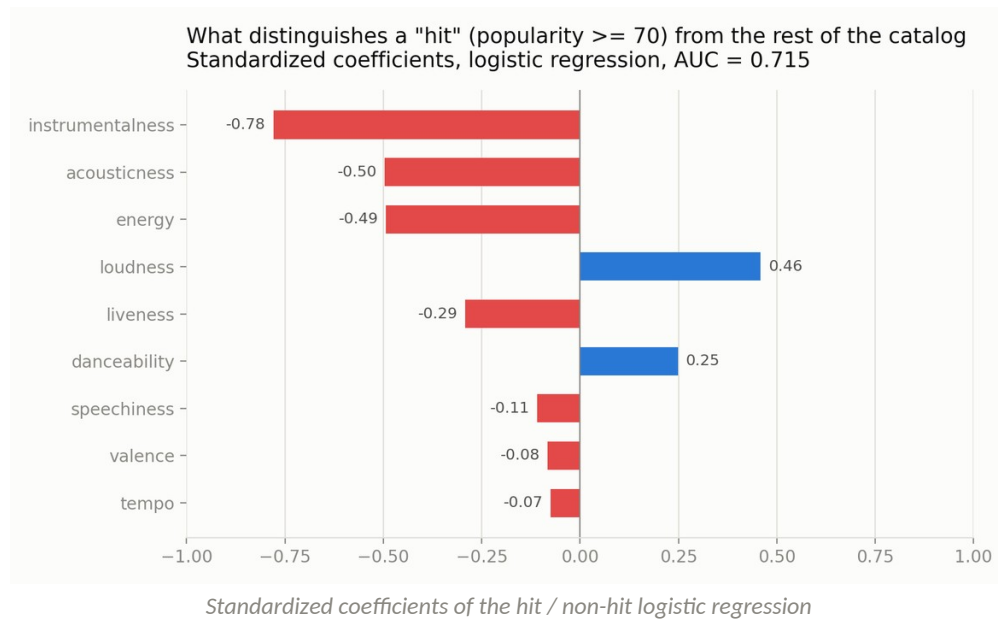


The initial hypothesis (“a larger, unbiased dataset would give a markedly higher R^2 ”) is NOT confirmed: R^2 remains very low on the 89,740 tracks (0.028), and is even slightly lower on the “hits only” subset of this new dataset (0.022) than on the original dataset (0.06). Adding extra variables (real explicit flag, key, mode, `time_signature`) changes almost nothing ($R^2 = 0.031$, i.e. +0.003).

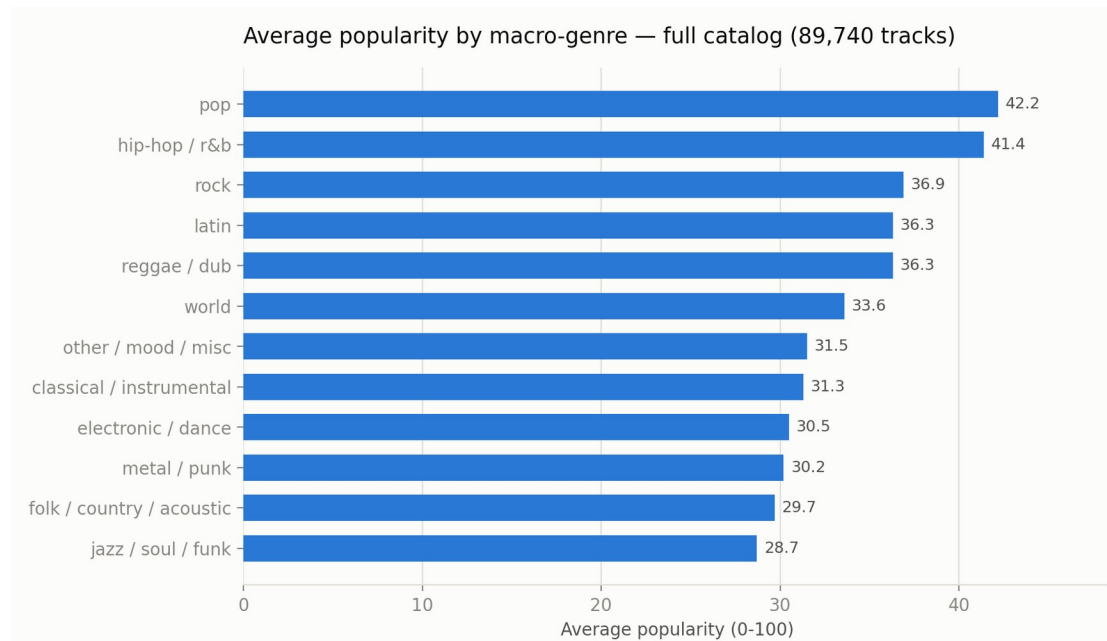
- **What this confirms, with far more robustness than before:** on a sample 89x larger and completely independent, the conclusion still holds: audio characteristics alone do not finely explain a track's popularity. This was not an artifact of the small original sample.

7.3 But a larger dataset answers questions the old one couldn't ask

Predicting the exact popularity score fails, but several different questions become possible once flops are included in the data:



- **Telling a hit apart from an ordinary track works much better than predicting a precise score:** a logistic regression on “popularity ≥ 70 or not” reaches an AUC of 0.715 (0.5 = chance, 1.0 = perfect). Hits have a recognizable profile: less instrumental, less acoustic, louder, more danceable; although, once inside that group, the audio says almost nothing about who will end up at a popularity of 75 versus 95.
- **Explicit content (a real metadata flag, more reliable than a proxy) is associated with higher popularity:** 36.9 on average versus 32.9 for non-explicit content. This is a statistical association observed across 89,740 tracks (comparison of means, confirmed in the multivariate regression with a positive coefficient); not proof of causation, and the gap remains modest. But it's a much more solid confirmation than in the original dataset, where only an unreliable proxy (speechiness, near-zero correlation) could be tested.
- **Key and mode (major/minor) were also tested:** tracks in minor mode have slightly higher popularity than tracks in major mode (33.64 vs 32.95), and the gap between the most popular key (E, 34.06) and the least popular (G, 32.61) is only 1.45 points on a 100-point scale. These gaps are statistically significant (t-test and ANOVA, $p < 0.0001$) only because the sample is huge (89,740 tracks); in practice, key and mode have almost no explanatory power over popularity. It's a good reminder that a “statistically significant” result isn't always a result that matters in practice.



Average popularity by macro-genre, full catalog (89,740 tracks)

- **The genre ranking changes completely once outside the “Top 100” bias:** across the full catalog, pop and hip-hop/r&b are the most popular macro-genres on average (42.2 and 41.4); jazz/soul/funk and folk/country are the least popular (28.7 and 29.7). This is a reading that was impossible with the original dataset, which only contained already-popular tracks.
- **The hit-conversion rate by genre is the most actionable conclusion of this extension:** hip-hop/r&b converts 10.5% of its tracks into hits (popularity ≥ 70), versus 8.0% for pop and barely 1.2% for classical/instrumental. Electronic/dance publishes the largest volume of tracks (16,639) but only converts at 3.5%; high volume doesn't mean a good success rate (SQL query Q5, `sql/extended_analysis_queries.sql`).

7.4 What can be cautiously concluded

The general conclusion holds: sound alone is not enough to explain a track's popularity, even on a very large and independent sample. One can legitimately wonder whether other factors, artist fame, promotional budget, playlist placement, play a bigger role, but this dataset measures none of these elements and so cannot confirm it; this is only a lead worth investigating, not a conclusion established here.

What a larger dataset did allow, very concretely, was answering questions the old dataset couldn't ask: the hit-conversion rate by genre (hip-hop/r&b and pop leading) didn't exist as a possible signal as long as the dataset contained no flops to calculate it.

In short, the answer to “would a larger dataset yield more conclusions” is yes, but not in the way initially assumed: more rows did not make the prediction model more accurate; however, a structurally different dataset, with flops, a real explicit-content flag, and enough volume to calculate conversion rates by genre, made it possible to answer business questions the old dataset couldn't ask.

8. Glossary

Definitions of the technical terms used in this report, for standalone reading.

- **Valence (perceived positivity):** a Spotify indicator (0 to 100 in this report) that measures the musical positivity conveyed by a track. A high score corresponds to a track that sounds cheerful or euphoric; a low score to a track that sounds sad or dark. This is a descriptor automatically computed by Spotify from the audio signal, not a listener survey.
- **Speechiness:** detects the presence of spoken rather than sung words. Close to 100 = an almost entirely spoken track (podcast, audiobook); between ~33 and 66 = a mix of singing/speech (rap); below 33 = “normal” music. This is NOT an indicator of explicit content (profanity). In the first version of the project, speechiness had been used as a proxy for “explicit content,” an approximation that turned out to be unreliable (see section 4.1 and section 7.3 for the real explicit flag).
- **Explicit content (explicit):** not to be confused with speechiness: this is metadata declared by the label or distributor when the track is uploaded (like an age rating), not something Spotify infers from the sound. It flags the presence of profanity, or references to sex, drugs, or violence in the lyrics. A simple Yes/No per track (true/false in the data), available only in the extended dataset from section 7.
- **Loudness:** the average sound volume of the track, in decibels (dB), generally between -60 and 0. The closer the value is to 0, the louder/more compressed the track is in the mix.
- **Acousticness:** a confidence score (0 to 100) that the track is acoustic, meaning produced without electronically amplified instruments or synthesizers. A high value = likely an acoustic version; a low value = more electronic production.
- **Key and mode (major/minor):** the track's musical key (C, D, E...) and mode (major or minor), in the classical musical sense. Analyzed in section 7.3: their effect on popularity is statistically measurable but very weak in practice (less than 1.5 points of gap out of 100).
- **R² (coefficient of determination):** measures the share of variance in the variable being predicted (here, popularity) explained by the model. It ranges from 0 to 1: 0 = the model explains nothing, 1 = the model explains all variation perfectly. An R² of 0.06 means the model explains 6% of the popularity differences between tracks; that's low.
- **AUC:** measures a classification model's ability (here, “hit or not”) to correctly distinguish the two classes. 0.5 = equivalent to chance, 1.0 = perfect distinction. An AUC of 0.715 is a decent result without being excellent.
- **Statistical vs. practical significance:** with a very large sample (89,740 tracks), even a tiny gap between two groups can become “statistically significant” ($p < 0.05$) in the mathematical sense, without that gap being large enough to matter in practice. Section 7.3 (key, mode) is a direct example of this.
- **Star schema modeling:** a way of organizing a database for analysis, with a central “fact” table (here, each track and its statistics) linked to “dimension” tables that describe context (the artist, the genre...). Drawn schematically, the fact table sits at the center with the dimensions around it, like the points of a star, hence the name. This is the standard model used by most BI and reporting tools.
- **What isn't measured in these datasets:** melody in the strict sense (the contour of the melodic line), lyrics, song structure (verse/chorus), and vocal quality. A real melody analysis would require transcribing the audio

signal (MIDI), outside the scope of this project, and access to certain Spotify endpoints has been restricted to new developer applications since late 2024.

Appendix: project files

- data/spotify_clean.csv, data/spotify.db: cleaned original dataset + SQL database (1000 tracks)
- data_v2/spotify_extended_clean.csv, data_v2/spotify_extended.db: cleaned extended dataset + SQL database (89,740 tracks)
- data/query_results/, data_v2/query_results/: results of all SQL queries
- python/01 to 05: pipeline for the original dataset; python/06 to 08: pipeline for the extended dataset
- sql/analysis_queries.sql, sql/extended_analysis_queries.sql: 16 commented queries in total
- dashboard/spotify_dashboard.html: standalone interactive dashboard
- Spotify_Project_Report.pdf (this document): methodology, insights, glossary